

# INTEGRATION TESTING

## 1. Able to read 2 or more configuration file names as command line parameters:

AIU/IT – 01, REQ1.1

```
aish@DESKTOP-V54RKL6:/mnt/c/Users/Aishwarya/OneDrive/Desktop/Automatic-Image-Update/make$ cd ../bin
aish@DESKTOP-V54RKL6:/mnt/c/Users/Aishwarya/OneDrive/Desktop/Automatic-Image-Update/bin$ ./aius_prg file.cfg file_kernel.KO application_file.apk file.BAK file.boot
```

If less than 2 file names are passed it throws invalid\_arguments exception and prints the message on the screen –

AIU/IT – 02, REQ 1.2

```
aish@DESKTOP-V54RKL6:/mnt/c/Users/Aishwarya/OneDrive/Desktop/Automatic-Image-Update/bin$ ./aius_prg file.cfg
Less number of arguments! Enter at least 2 filenames.
aish@DESKTOP-V54RKL6:/mnt/c/Users/Aishwarya/OneDrive/Desktop/Automatic-Image-Update/bin$ █
```

2) Each configuration file will consists of header and data.

Header:

Device Id: <Unique 6 Digit alphanumeric Id>

Magic No:<Unique Device specific 16 Digit number>

Device Name: <255 char device name>

Checksum:Checksum of suitable length to ensure data integrity

Base Address:

BOOT IMAGE:start\_addr=<32 bit/64 w.r.t start address>, size=<size>, ver=<version string>

---- AIU/IT-03, REQ2

```

C:\aish@DESKTOP-V54RKLK: /mnt/c/Users/Aishwarya/OneDrive/Desktop/Automatic-Image-Update/bin

===== Header details of Configuration files =====

Device id: GF156U
Magic No: 1234567898765432
Device Name: Configuration_File_Image
Base Address: 0x11011100
Checksum: 00001000
bootImageAddress: 0x3E00    bootImageSize: 64    bootImageVersion: 10.6.16299.15

File Type: file system
-----
Device id: 1EBH56
Magic No: 1234567898765432
Device Name: Kernal_Image
Base Address: 0x10111010
Checksum: 00000000
bootImageAddress: 0x3E00    bootImageSize: 64    bootImageVersion: 10.1.16299.15

File Type: kernel module
-----
Device id: PQ1578
Magic No: 257832517578935
Device Name: Application_Image
Base Address: 0x01110100
Checksum: 9F097AAEEF57D6A165D95BE1590662626802
bootImageAddress: 0x3C00    bootImageSize: 64    bootImageVersion: 10.2.16299.15

File Type: application
-----

```

3) Source files should get picked from default path

Source files are present in the bin folder where the .exe file is present

is PC > Desktop > Automatic-Image-Update > bin

| Name                     | Date modified    | Type                  | Size   |
|--------------------------|------------------|-----------------------|--------|
| aius_prg                 | 11-04-2022 21:33 | File                  | 383 KB |
| application_file.apk     | 07-04-2022 16:06 | APK File              | 1 KB   |
| file.BAK                 | 09-04-2022 19:58 | BAK File              | 1 KB   |
| file.boot                | 09-04-2022 19:39 | BOOT File             | 1 KB   |
| file.cfg                 | 09-04-2022 19:39 | CFG File              | 1 KB   |
| file_kernal.KO           | 09-04-2022 19:39 | KO File               | 1 KB   |
| framework_file.FRAMEWORK | 09-04-2022 19:40 | FRAMEWORK File        | 1 KB   |
| library_file.DLL         | 09-04-2022 19:41 | Application extens... | 1 KB   |

4) A memory mapped region to simulate Target device --- AIU/IT-04 , REQ3

```
Enter your choice: 2

===== Device details =====
Start of device :      0x7fbbb0810010
Application :         0x7fbbb1c10025 :
boot image :         0x7fbbb081002a :
libraries :          0x7fbbb3010020 :
framework level :    0x7fbbb441001b :
file system :        0x7fbbb5810016 :
kernel module :      0x7fbbb6c10011 :
End Address :        0x7fbbb801000f
Total Length : 125829120 bytes
```

5) Should be able to query and get the current device information including details about all images

- Option to get the Header details of all configuration files -- -- AIU/IT-05, REQ4.1

```
===== AUTOMATIC IMAGE UPDATE SOFTWARE =====

1. Save and display the Header details of files entered
2. Display Device Details
3. To confirm file updation
4. Exit

Enter your choice: 1

===== Header details of Configuration files =====

Device id: GF156U
Magic No: 1234567898765432
Device Name: Configuration_File_Image
Base Address: 0x11011100
Checksum: 00001000
bootImageAddress: 0x3E00    bootImageSize: 64    bootImageVersion: 10.6.16299.15

File Type: file system
-----
Device id: 1EBH56
Magic No: 1234567898765432
Device Name: Kernal_Image
Base Address: 0x10111010
Checksum: 00000000
bootImageAddress: 0x3E00    bootImageSize: 64    bootImageVersion: 10.1.16299.15

File Type: kernel module
-----
```

- Option to display Device details and its partition -- AIU/IT-06, REQ4.2

```

Enter your choice: 2

===== Device details =====
Start of device :      0x7fbbb0810010
application :         0x7fbbb1c10025 :
boot image :         0x7fbbb081002a :
libraries :          0x7fbbb3010020 :
framework level :    0x7fbbb441001b :
file system :        0x7fbbb5810016 :
kernel module :      0x7fbbb6c10011 :
End Address :        0x7fbbb801000f
Total Length : 125829120 bytes

```

- Option to update the files --- REQ4.3

```

1. Save and display the Header details of files entered
2. Display Device Details
3. To confirm file updation
4. Exit

Enter your choice: 3
Device not validated
Validated device

00000000

Updation complete for file_kernal.KO

```

6) In case of failure to upload all the image files, it should revert to original state. --- AIU/IT-07, REQ5.1

```
Enter your choice: 3
Device not validated
Validated device

00000000

Updation complete for file_kernal.KO

Device not validated
Validated device

00000000

Updation failed for file.BAK

Validated device

00000000

Updation failed for file.boot
```

Updation failed due to  
invalid header details  
or in valid checksum

Updation failed due to  
invalid header details  
or in valid checksum

Device details reverted back to it's original state :

```
Updation failed for file.boot

1. Save and display the Header details of files entered
2. Display Device Details
3. To confirm file updation
4. Exit

Enter your choice: 2

===== Device details =====
Start of device :      0x7f7bf4420010
application :          0x7f7bf5820025 :
boot image :           0x7f7bf442002a :
libraries :            0x7f7bf6c20020 :
framework level :      0x7f7bf802001b :
file system :           0x7f7bf9420016 :
kernel module :         0x7f7bfa820011 :
End Address :           0x7f7bfb02000f
Total Length : 125829120 bytes
```

When all the configuration files are validated: --- AIU/IT – 08, REQ5.2

```

aish@DESKTOP-V54RKLK: /mnt/c/Users/Aishwarya/OneDrive/Desktop/Automatic-Image-Update/bin
Enter your choice: 3

Validated device
00000000

Updation complete for file_kernal.KO

Validated device
00000000

Updation complete for file.BAK
```

Device details are updated here in the heap memory

```

aish@DESKTOP-V54RKLK: /mnt/c/Users/Aishwarya/OneDrive/Desktop/Automatic-Image-Update/bin
Enter your choice: 2

===== Device details =====
Start of device :      0x7f3f22870010
application :      0x7f3f23c70025 :
boot image :      0x7f3f2287002a :
libraries :      0x7f3f25070020 :
framework level :      0x7f3f2647001b :
file system :      0x7f3f27870016 : 1EBH56
1234567898765432
Filesystem_Image
00000000
0x11011110
0x3F00 64 10.5.16299.15

kernel module :      0x7f3f28c70011 : 1EBH56
1234567898765432
Kernal_Image
00000000
0x10111010
0x3E00 64 10.1.16299.15
This is an instance of data
ther are more data

End Address :      0x7f3f2a07000f
Total Length : 125829120 bytes

1. Save and display the Header details of files entered
2. Display Device Details
3. To confirm file updation
4. Exit

Enter your choice: 4

Exiting...
Update finished

===== CLOSING AUTOMATIC IMAGE UPDATE SOFTWARE =====
```